



EE 354/CE 361/MATH 310 -
Introduction to Probability
and Statistics
Spring 2026





EE 354/CE 361/MATH 310 L4 section (Spring 2026)

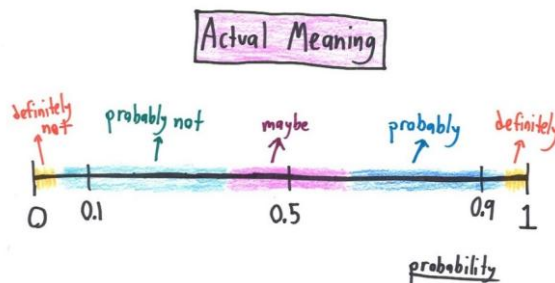
Introduction to Probability and Statistics -

Aamir Hasan

Week 11 - Lecture No 21 – March 25, 2026

❖ Online class (to be strictly adhered)

1. Please ensure audio and video is working (see below)
2. I will place questions to the students and not responding w/ audio & video would be used for participation & attendance.
3. Cameras to remain switched 'ON' during the entire lecture (else attendance would not be marked)
4. In cases where I am unable to ask questions to some individuals, attendance would be taken sometime in between the lecture.



Announcements!

1. Office hours (post Ramadan) – Tuesday & Thursday 12:00 to 1:00 PM
2. Quiz No 06 – April 01, 2026

Agenda for today

1. Unit 5 – Continuous Random Variable

Probability Density Functions (PDFs):

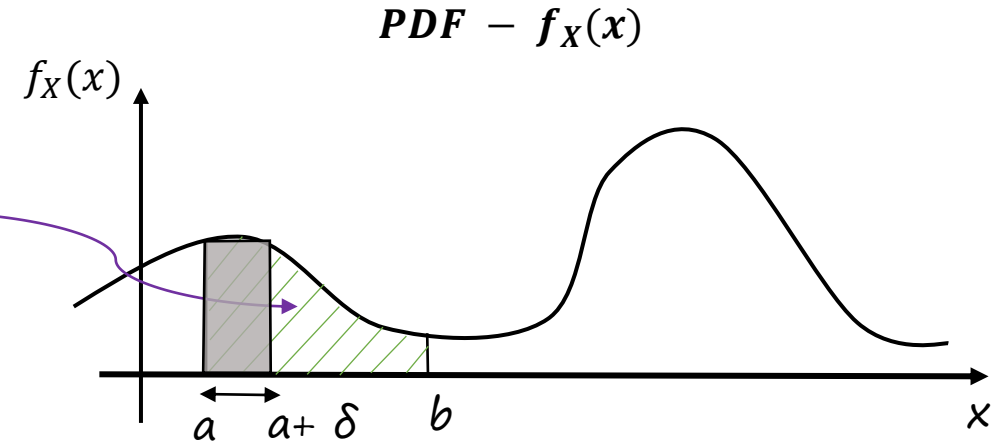
notion of $f_X(a)$

$\delta > 0$, small (approaching zero)



$$\mathbf{P}(a \leq X \leq a + \delta) \approx f_X(a)\delta$$

$$\mathbf{P}(a \leq X \leq b)$$



$$\mathbf{P}(a \leq X \leq b) = \int_a^b f_X(x) dx$$

- $\int_{-\infty}^{\infty} f_X(x) dx = 1$
- $f_X(x) \geq 0$

Red arrow pointing right

$$\mathbf{P}(a \leq X \leq b) = \cancel{\mathbf{P}(x = a)} + \cancel{\mathbf{P}(x = b)} + \mathbf{P}(a < X < b)$$

$$\mathbf{P}(a \leq X \leq b) = \mathbf{P}(a < X < b)$$

Properties of Expectations

- If $X \geq 0$, then $E[X] \geq 0$
- If $a \leq X \leq b$, then $a \leq E[X] \leq b$
- Expected value rule:

$$E[g(X)] = \sum_x g(x)p_X(x)$$

$$\rightarrow E[X^2] = \int_{-\infty}^{\infty} x^2 f_X(x) dx$$

- Linearity

$$E[g(X)] = \int_{-\infty}^{\infty} g(x)f_X(x) dx$$

$$E[aX + b] = aE[X] + b$$

Variance and its properties

Definition of Variance: $\text{var}(X) = E[(X - \mu)^2]$

$$\mu = E[X]$$

→ • Calculation using the expected value rule: $E[g(X)] = \int_{-\infty}^{\infty} g(X) f_X(x) dx$

$$\Rightarrow \text{var}(X) = \int_{-\infty}^{\infty} (X - \mu)^2 f_X(x) dx \quad g(X) = (X - \mu)^2$$

→ $\text{var}(aX + b) = a^2 \text{var}(X)$

→ Standard Deviation: $\sigma_{(X)} = \sqrt{\text{var}(X)}$

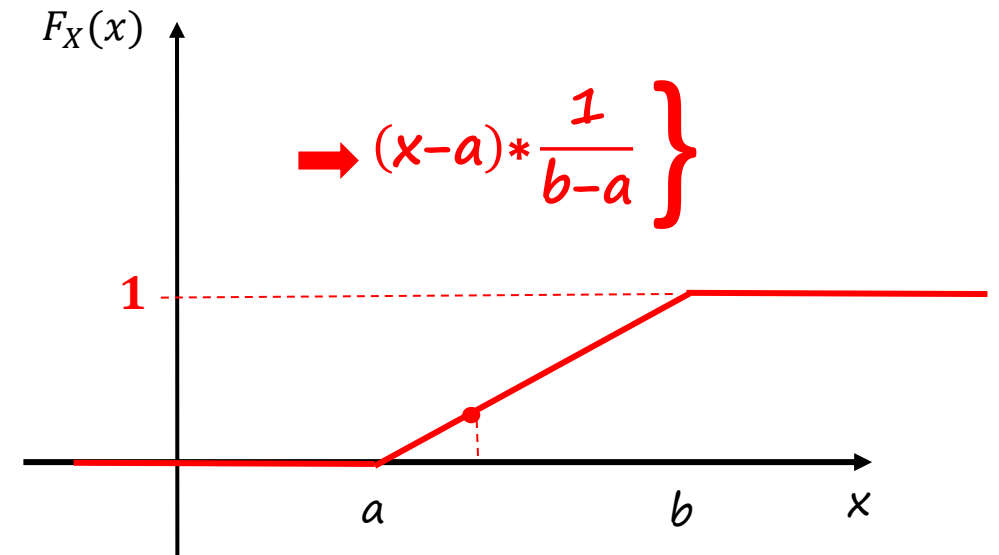
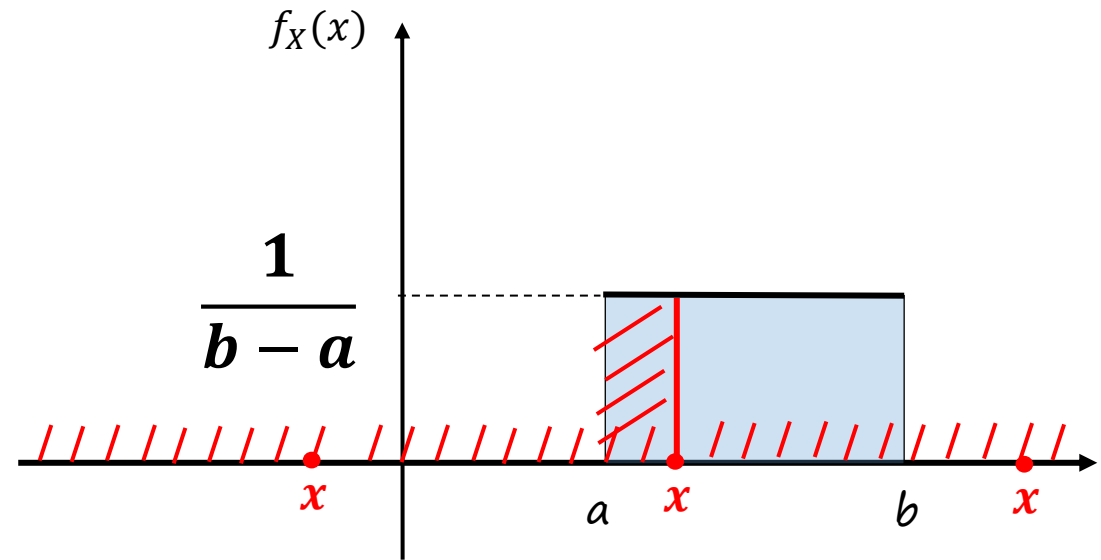
→ A useful formula: $E[X^2] - (E[X])^2$

Cumulative distribution function (CDF):

CDF definition: $F_X(x) = P(X \leq x)$

1. Continuous random variables:

$$F_X(x) = p(X \leq x) = \int_{-\infty}^x f_X(t) dt$$



Cumulative distribution function (CDF):

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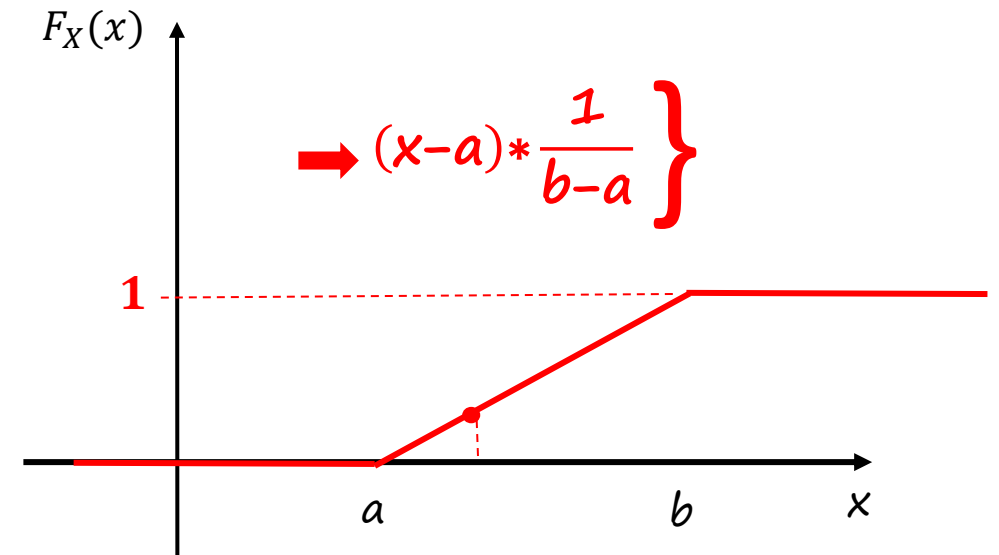
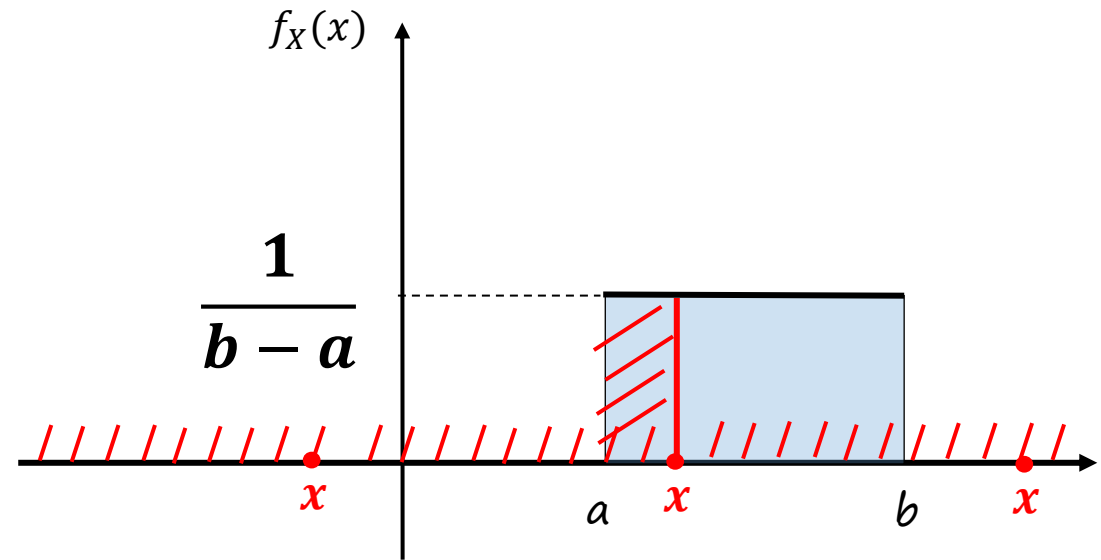


$$\rightarrow P(X \leq 4) = P(X \leq 3) + P(3 < X \leq 4)$$

$$\rightarrow F_X(4) = F_X(3) + P(3 < X \leq 4)$$

$$\rightarrow P(3 \leq X \leq 4) = F_X(4) - F_X(3)$$

$$\rightarrow \boxed{\frac{dF_X}{dx}(x) = f_X(x)}$$



Cumulative distribution function (CDF)

CDF definition: $F_X(x) = P(X \leq x)$

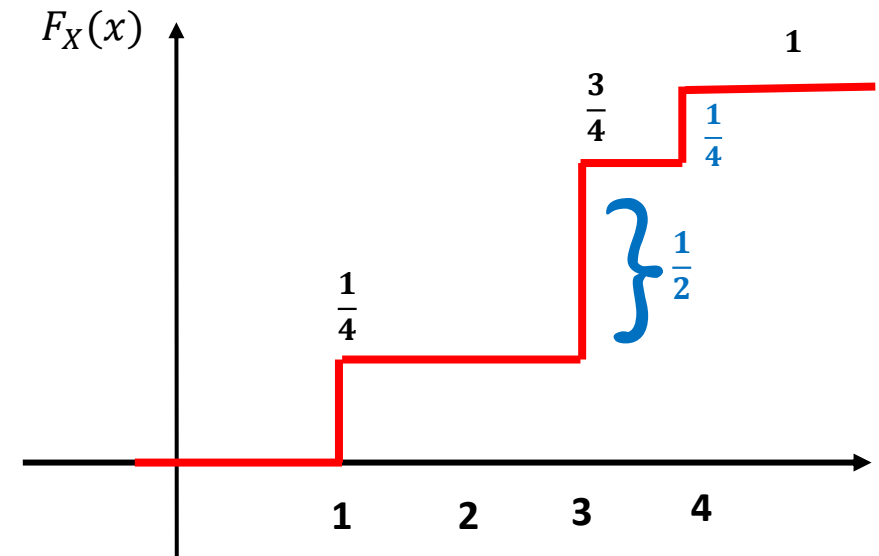
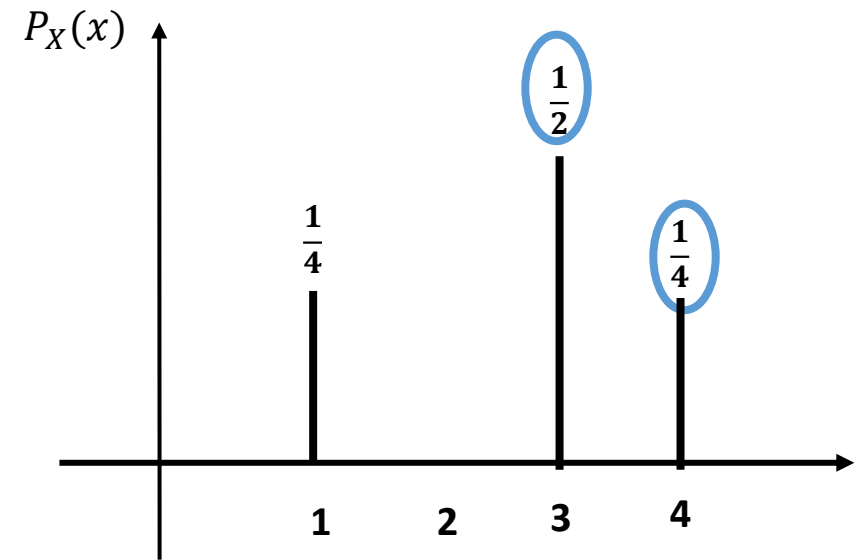
2. Discrete random variables:

$$F_X(x) = p(X \leq x) = \sum_{k \leq x} p_X(k)$$

$$p(X \leq 1) = 1/4$$

$$p(X \leq 3) = 3/4$$

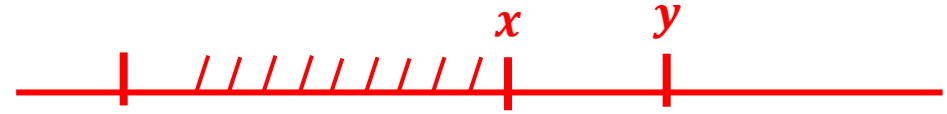
$$p(X \leq 4) = 1$$



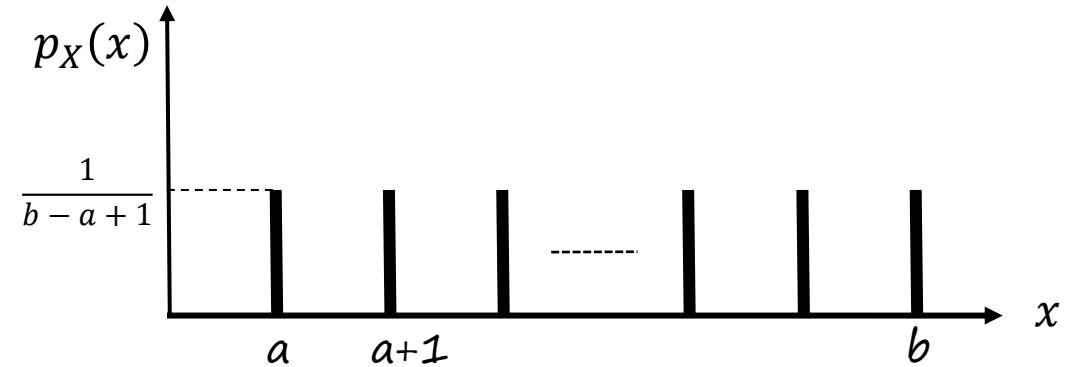
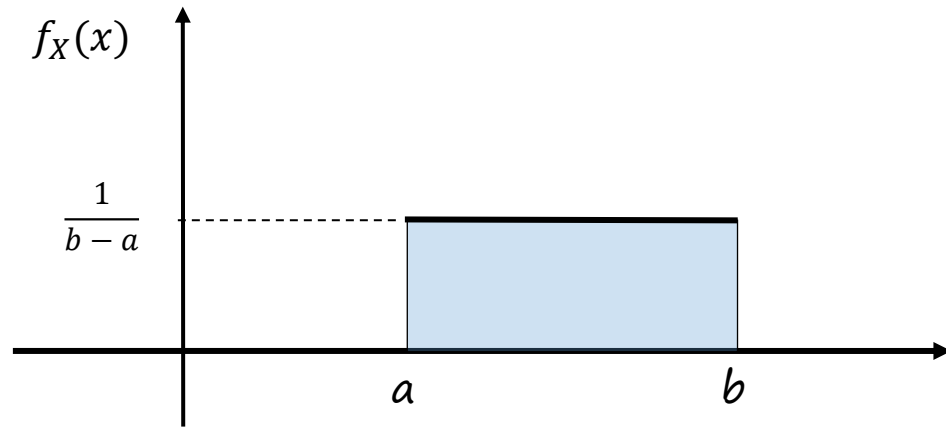
General CDF properties

CDF definition: $F_X(x) = P(X \leq x)$

- Non-decreasing $\Rightarrow$ if $y \geq x$ then $F_X(y) \geq F_X(x)$
- $F_X(x)$ tends to 1, as $x \rightarrow \infty$
- $F_X(x)$ tends to 0, as $x \rightarrow -\infty$



Continuous Uniform Random Variable; parameters a, b



$$E[X] = \int_{-\infty}^{\infty} x f_X(x) dx$$

→
$$= \int_a^b x \frac{1}{b-a} dx = \frac{a+b}{2}$$

→
$$E[X^2] = \int_a^b x^2 \frac{1}{b-a} dx = \frac{1}{b-a} \left(\frac{b^3}{3} - \frac{a^3}{3} \right)$$

→
$$\text{var}(X) = E[X^2] - (E[X])^2 = \frac{(b-a)^2}{12} \quad \rightarrow \quad \sigma = \frac{b-a}{\sqrt{12}}$$

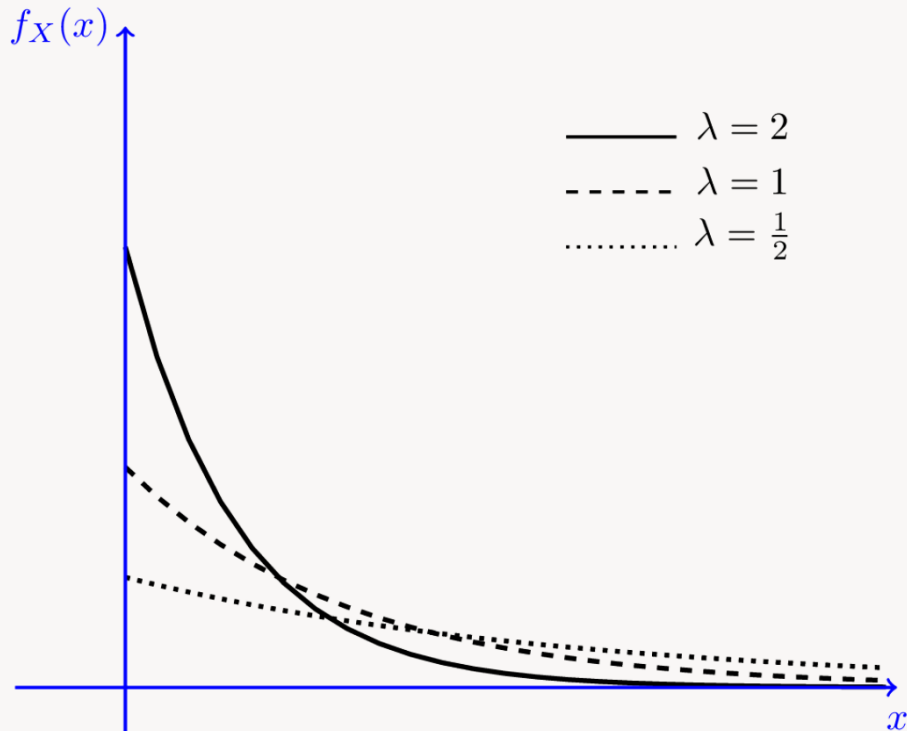
$$E[X] = \frac{a+b}{2}$$

$$\text{var}(X) = \frac{1}{12} (b-a)(b-a+2)$$

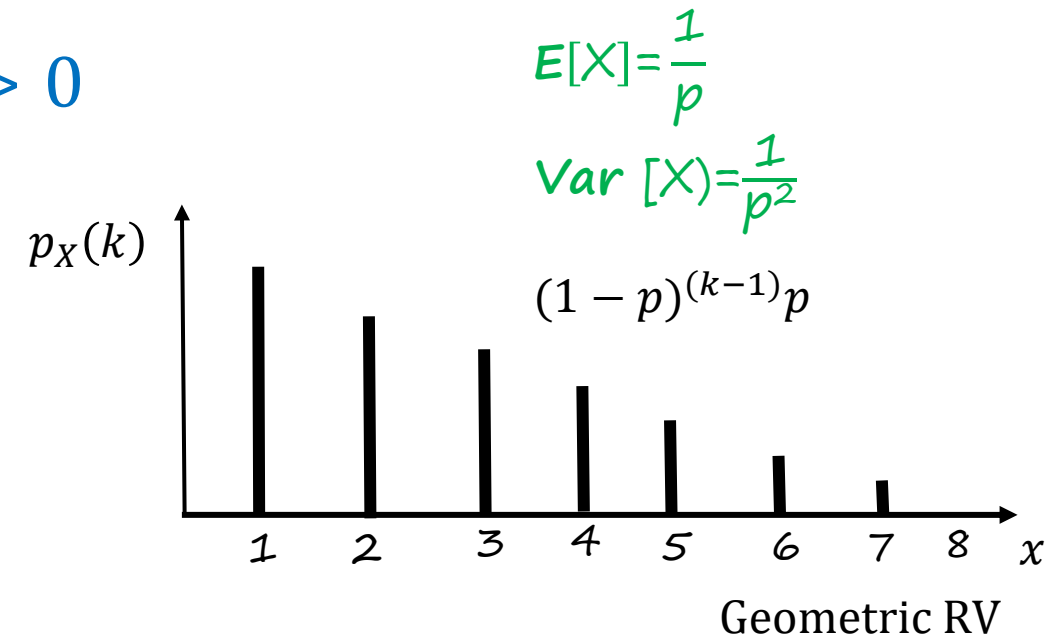
Exponential Random Variable; parameter $\lambda > 0$

$$f_X(x) = \begin{cases} \lambda e^{-\lambda x}, & x \geq 0 \\ 0, & x < 0 \end{cases}$$

Here $\lambda > 0$, is the parameter of the distribution, often called the *rate parameter*.



PDF of the exponential random variable



Exponential distribution is the probability distribution of the distance between events in a Poisson point process, i.e., a process in which events occur continuously and independently **at a constant average rate**; the distance parameter could be any meaningful mono-dimensional measure of the process, such as time between production errors, time of arrival, waiting times etc.

An important property - exponential distribution is **memoryless**

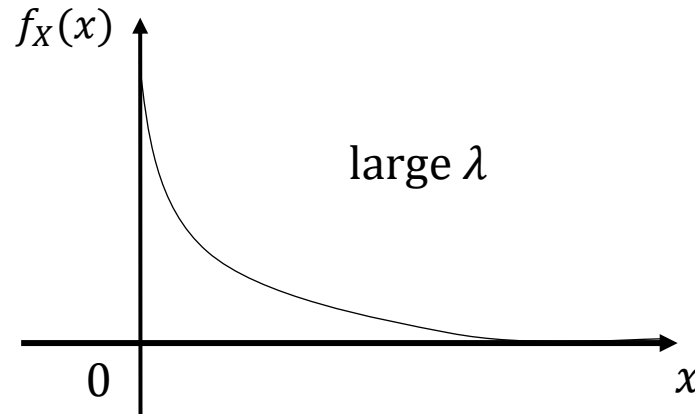
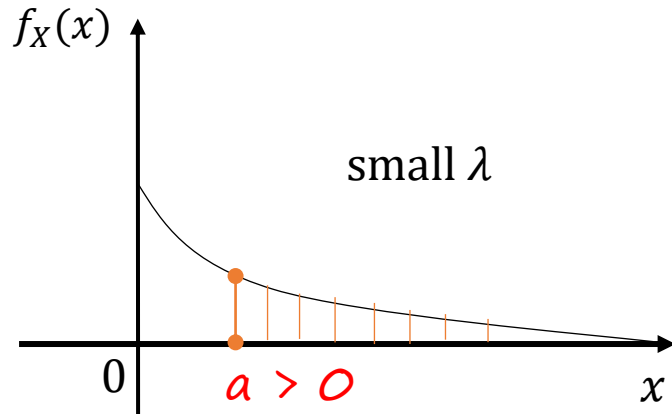
If X is exponential with parameter $\lambda > 0$, then X is a *memoryless* random variable, that is

$$P(X > x + a \mid X > a) = P(X > x), \quad \text{for } a, x \geq 0.$$

Waiting time until arrival of a customer - the memoryless property means that it does not matter how long you have waited so far. If you have not observed a customer until time a , the distribution of waiting time (from time a) until the next customer is the same as when you started at time zero.

Exponential Random Variable; parameter $\lambda > 0$

$$f_X(x) = \begin{cases} \lambda e^{-\lambda x}, & x \geq 0 \\ 0, & x < 0 \end{cases}$$



$$\rightarrow E[X] = \int_0^{\infty} x \lambda e^{-\lambda x} dx = \frac{1}{\lambda}$$

$$\rightarrow E[X^2] = \int_0^{\infty} x^2 \lambda e^{-\lambda x} dx = \frac{2}{\lambda^2}$$

$$\rightarrow \text{var}(X) = E[X^2] - (E[X])^2 = \frac{2}{\lambda^2} - \frac{1}{\lambda^2} = \frac{1}{\lambda^2}$$

$$P(X \geq a) = ?$$

$$P(X \geq a) = \int_a^{+\infty} \lambda e^{-\lambda x} dx$$

$$= \lambda \left(\frac{-1}{\lambda} \right) e^{-\lambda x} \Big|_a^{\infty}$$

$$= -e^{-\lambda \infty} + e^{-\lambda a} = e^{-\lambda a}$$

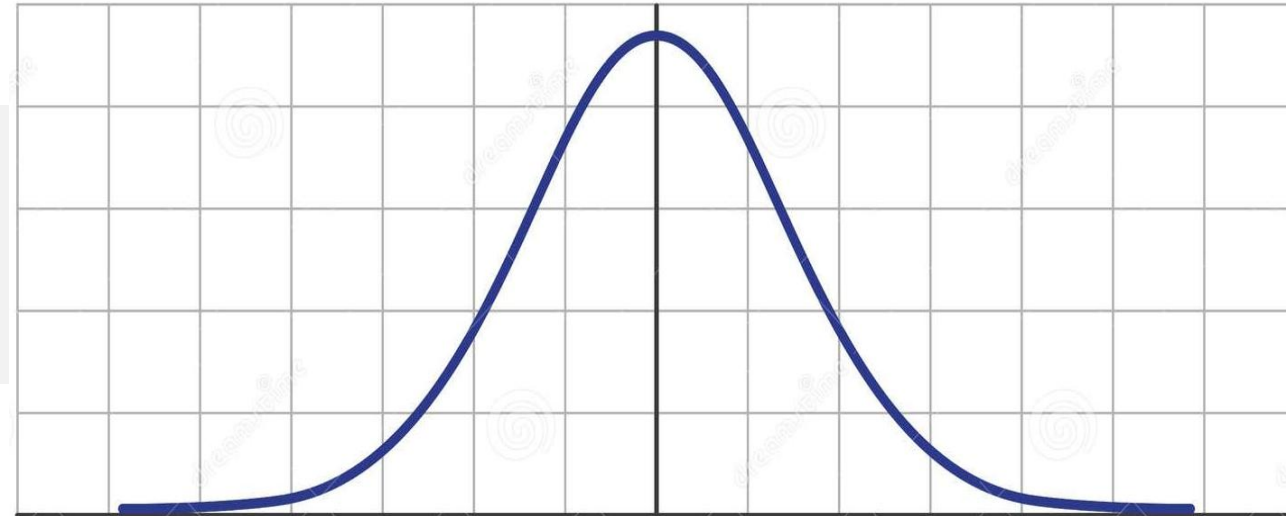
Normal (Gaussian Random Variables) - *Why They Matter in the Real World*

Gaussian or Normal random variables - one of the most fundamental patterns of randomness in nature & engineering.

The Gaussian distribution shows up because of the **Central Limit Theorem**: when many small, independent random effects combine, their sum tends to look Gaussian, regardless of the original distribution. This makes it a natural model for complex, real-world processes.

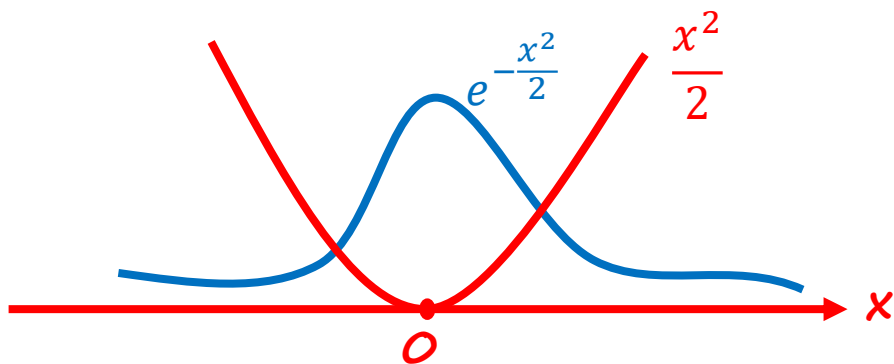
Everyday and Engineering Applications: **Noise in Communication Systems (AWGN)**, **Electronics and Circuits** (*Thermal or shot noise*), **Image and Signal Processing** (*Understanding the Gaussian helps you build filters like the Wiener filter or Kalman filter, widely used in control, robotics, and tracking systems*), **Natural and Human Phenomena** (*Human height, measurement errors, and even exam scores often approximate Gaussian distributions.*) - **that's why the "bell curve" appears so often in statistics and performance evaluation.**

**The Gaussian RV is not just a mathematical curiosity
- it is the language of randomness that engineers
use to model, analyze, and design systems that
work reliably in the noisy, uncertain world.**



Normal (Gaussian Random Variables)

- Standard normal $N(0,1)$: $f_X(x) = \frac{1}{\sqrt{2\pi}} e^{-\frac{x^2}{2}}$



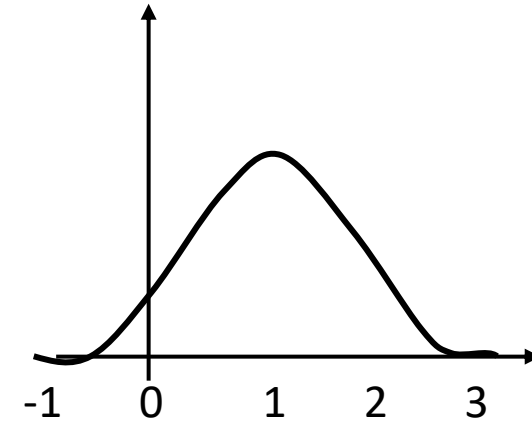
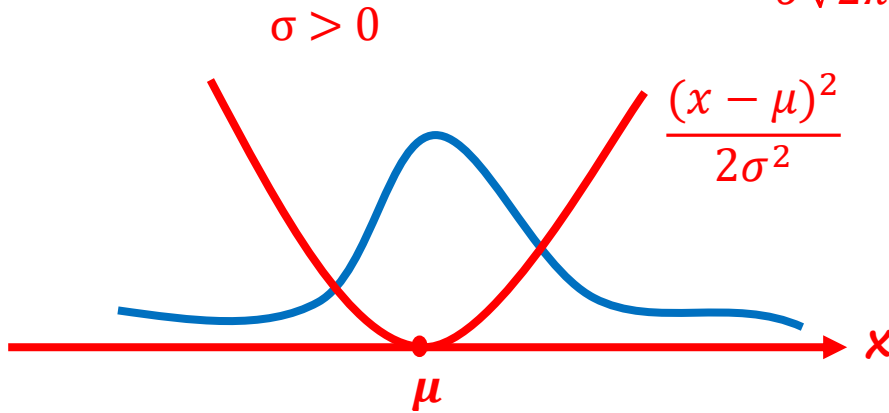
Calculus:

$$\int_{-\infty}^{+\infty} e^{-\frac{x^2}{2}} dx = \sqrt{2\pi}$$

- $E[X] = 0$
- $\text{Var}(X) = 1$

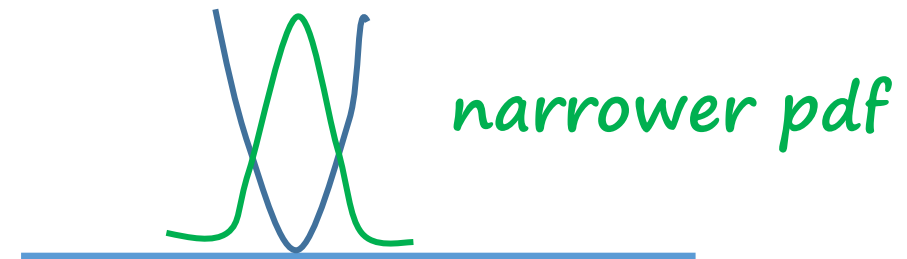
General normal (Gaussian) Random Variables

- General normal $N(\mu, \sigma^2)$: $f_X(x) = \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{(x-\mu)^2}{2\sigma^2}}$



σ small

- $E[X] = \mu$
- $\text{Var}(X) = \sigma^2$



Linear functions of a normal random variable

- Let $Y = aX + b$ where $X \sim N(\mu, \sigma_x^2)$

$$E[Y] = a\mu + b$$

$$\text{Var}[Y] = a^2\sigma_x^2$$

- Fact (will prove later in this course):

$$Y \sim N(a\mu + b, a^2\sigma_x^2)$$

- Special case: $a = 0$?

$$\begin{array}{l} Y = b \quad \text{discrete} \\ \uparrow \\ N(b, 0) \end{array}$$

Example – Calculating Probability for a Standard Normal (Gaussian Random Variables)

$$X \sim N(0, 1)$$

1. find $P(X \geq 0)$?

$$P(X \geq 0) = 1/2$$

2. find $P(X \geq 10)$?

$$= \int_{10}^{\infty} \frac{1}{\sqrt{2\pi}} e^{-\frac{x^2}{2}} dx$$

No closed form available

3. How about $P(X \leq x)$?

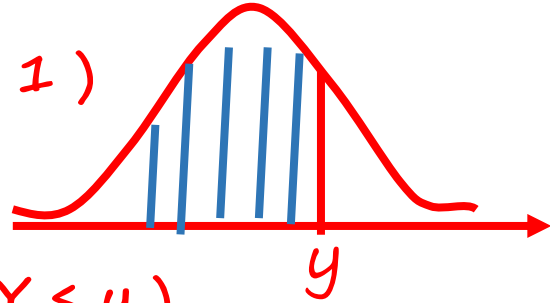
$$= \int_{-\infty}^x \frac{1}{\sqrt{2\pi}} e^{-\frac{x^2}{2}} dx$$

No closed form CDF

Standard Normal Tables

- No closed form available for CDF
- But we have tables, for the standard normal

→ $Y = N(0, 1)$

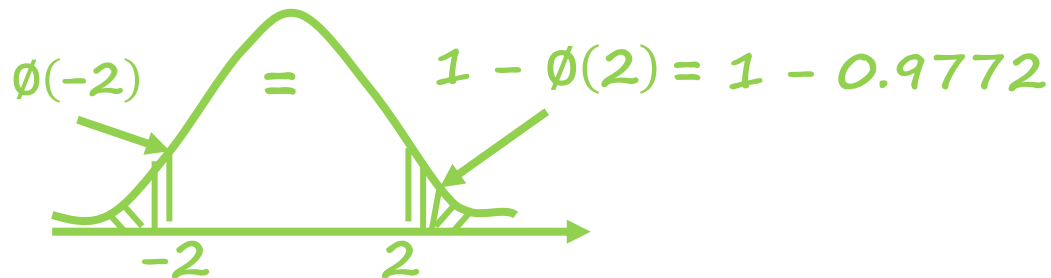


→ $\Phi(y) = F_Y(y) = P(Y \leq y)$

→ $\Phi(0) = P(Y \leq 0) = 0.5$

→ $\Phi(1.16) = 0.8770$

→ $\Phi(2.9) = 0.9981$

[illegible]